

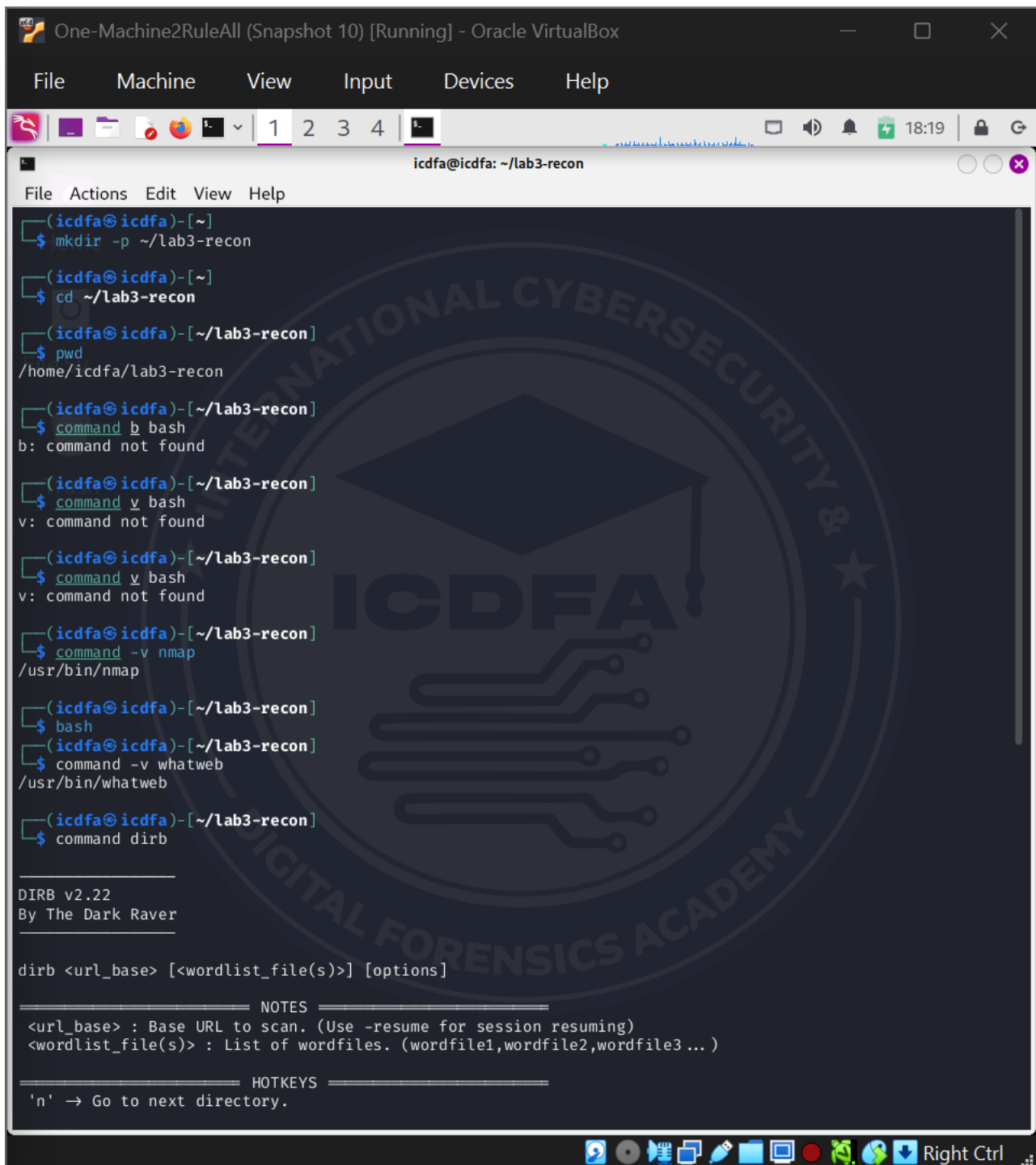
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Registration number: C11/26/FCDF/17167

Linux distribution / version: Kali GNU/ 2025.1

VM hostname: Oracle VirtualBox

Part 2



```
One-Machine2RuleAll (Snapshot 10) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

icdfa@icdfa: ~/lab3-recon
File Actions Edit View Help

(icdfa@icdfa)-[~]
$ mkdir -p ~/lab3-recon

(icdfa@icdfa)-[~]
$ cd ~/lab3-recon

(icdfa@icdfa)-[~/lab3-recon]
$ pwd
/home/icdfa/lab3-recon

(icdfa@icdfa)-[~/lab3-recon]
$ command b bash
b: command not found

(icdfa@icdfa)-[~/lab3-recon]
$ command v bash
v: command not found

(icdfa@icdfa)-[~/lab3-recon]
$ command v bash
v: command not found

(icdfa@icdfa)-[~/lab3-recon]
$ command -v nmap
/usr/bin/nmap

(icdfa@icdfa)-[~/lab3-recon]
$ bash

(icdfa@icdfa)-[~/lab3-recon]
$ command -v whatweb
/usr/bin/whatweb

(icdfa@icdfa)-[~/lab3-recon]
$ command dirb

DIRB v2.22
By The Dark Raver

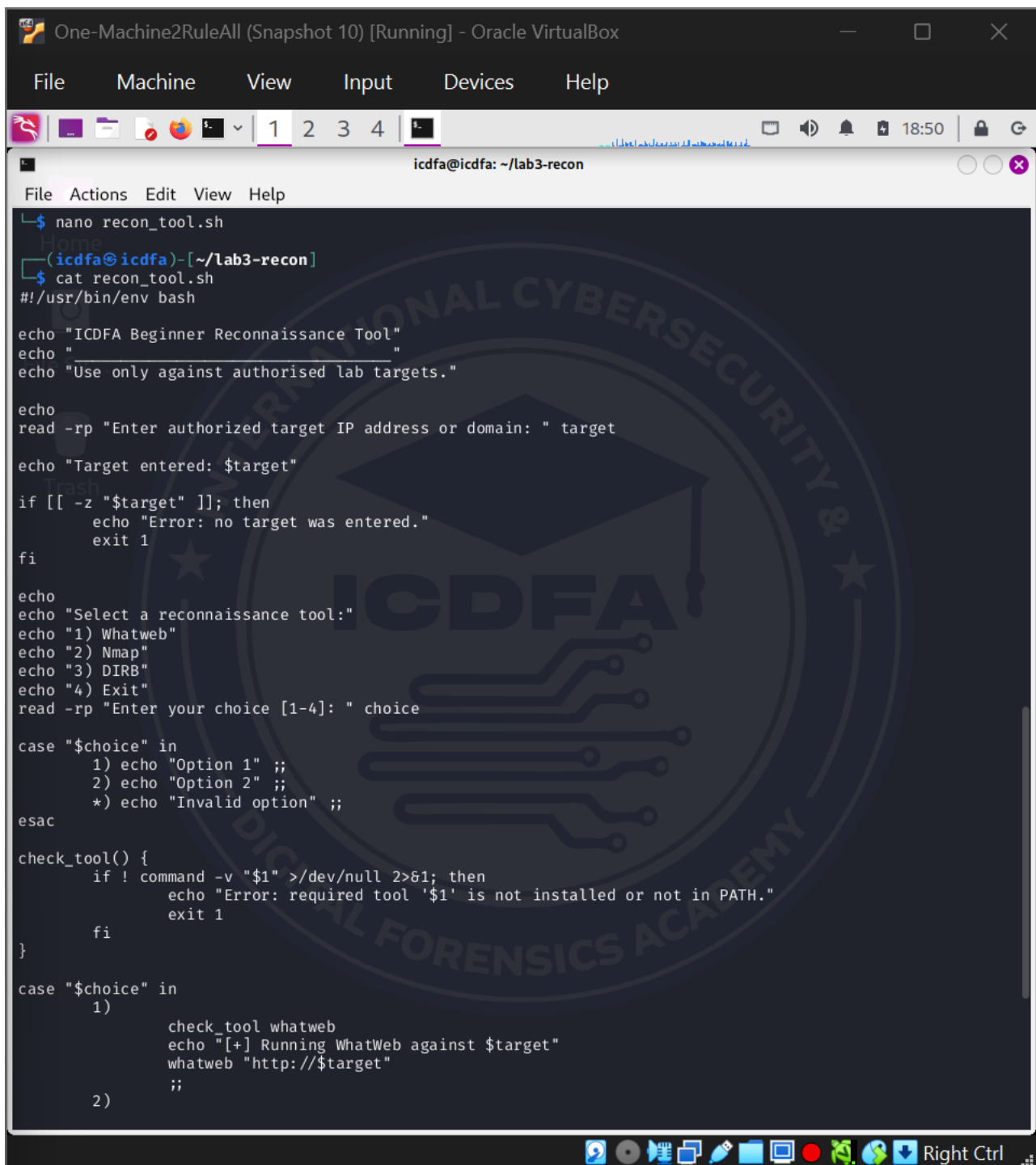
dirb <url_base> [<wordlist_file(s)>] [options]

===== NOTES =====
<url_base> : Base URL to scan. (Use -resume for session resuming)
<wordlist_file(s)> : List of wordfiles. (wordfile1,wordfile2,wordfile3 ...)

===== HOTKEYS =====
'n' -> Go to next directory.
```



Part 5



The screenshot shows a VirtualBox window titled "One-Machine2RuleAll (Snapshot 10) [Running] - Oracle VirtualBox". Inside the window is a Linux terminal window titled "icdfa@icdfa: ~/lab3-recon". The terminal displays the contents of a shell script named "recon_tool.sh" created with nano. The script is a Bash script that prompts the user for a target IP address or domain, validates the input, and then prompts for a choice of reconnaissance tool (1) Whatweb, (2) Nmap, (3) DIRB, or (4) Exit. It includes a function "check_tool" to verify if the selected tool is installed and in the PATH. The script is currently showing the first part of the case statement for choice 1, which uses "whatweb" to scan the target.

```
icdfa@icdfa: ~/lab3-recon
File Actions Edit View Help
$ nano recon_tool.sh
(icdfa@icdfa)-[~/lab3-recon]
$ cat recon_tool.sh
#!/usr/bin/env bash

echo "ICDFA Beginner Reconnaissance Tool"
echo "_____"
echo "Use only against authorised lab targets."

echo
read -rp "Enter authorized target IP address or domain: " target

echo "Target entered: $target"

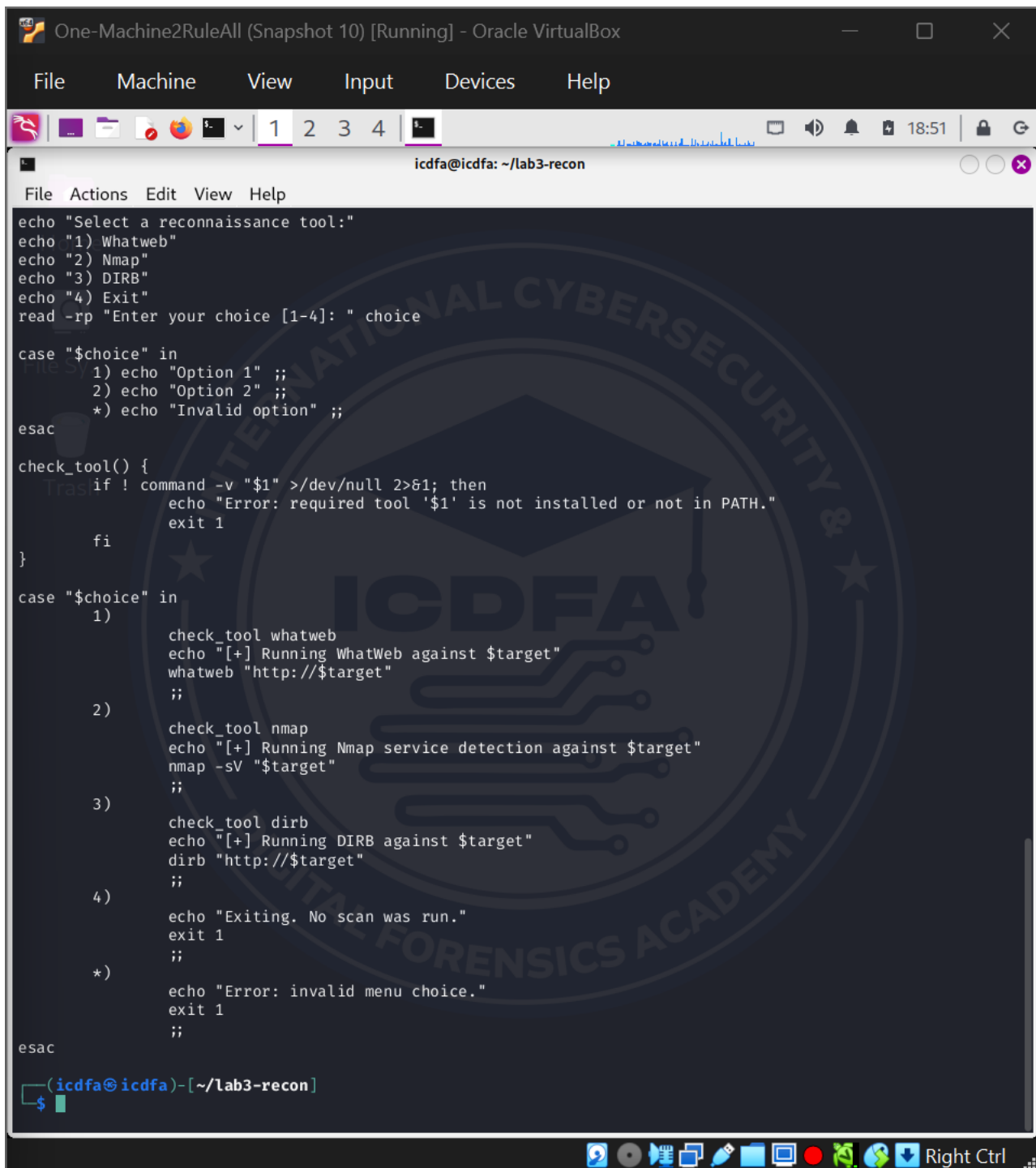
if [[ -z "$target" ]]; then
    echo "Error: no target was entered."
    exit 1
fi

echo
echo "Select a reconnaissance tool:"
echo "1) Whatweb"
echo "2) Nmap"
echo "3) DIRB"
echo "4) Exit"
read -rp "Enter your choice [1-4]: " choice

case "$choice" in
    1) echo "Option 1" ;;
    2) echo "Option 2" ;;
    *) echo "Invalid option" ;;
esac

check_tool() {
    if ! command -v "$1" >/dev/null 2>&1; then
        echo "Error: required tool '$1' is not installed or not in PATH."
        exit 1
    fi
}

case "$choice" in
    1)
        check_tool whatweb
        echo "[+] Running WhatWeb against $target"
        whatweb "http://$target"
        ;;
    2)
        ;;
```



One-Machine2RuleAll (Snapshot 10) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

icdfa@icdfa: ~/lab3-recon

```
File Actions Edit View Help
case "$choice" in
1) echo "Option 1" ;;
2) echo "Option 2" ;;
*) echo "Invalid option" ;;
esac

check_tool() {
if ! command -v "$1" >/dev/null 2>&1; then
echo "Error: required tool '$1' is not installed or not in PATH."
exit 1
fi
}

case "$choice" in
1)
check_tool whatweb
echo "[+] Running WhatWeb against $target"
whatweb "http://$target"
;;
2)
check_tool nmap
echo "[+] Running Nmap service detection against $target"
nmap -sV "$target"
;;
3)
check_tool dirb
echo "[+] Running DIRB against $target"
dirb "http://$target"
;;
4)
echo "Exiting. No scan was run."
exit 1
;;
*)
echo "Error: invalid menu choice."
exit 1
;;
esac

(icdfa@icdfa)-[~/lab3-recon]
$ chmod +x recon_tool.sh

(icdfa@icdfa)-[~/lab3-recon]
$ ls -l recon_tool.sh
-rwxrwxr-x 1 icdfa icdfa 1161 Sep  6 18:49 recon_tool.sh

(icdfa@icdfa)-[~/lab3-recon]
$
```

Right Ctrl



Part 6



```
One-Machine2RuleAll (Snapshot 10) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

icdfa@icdfa: ~/lab3-recon
File Actions Edit View Help

3) DIRB
4) Exit
Enter your choice [1-4]: 1
Option 1
[+] Running WhatWeb against 10.0.2.3
ERROR Opening: http://10.0.2.3 - execution expired
^[[A^[[B^[[A
(icdfa@icdfa)-[~/lab3-recon]
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool

Use only against authorised lab targets.

Enter authorized target IP address or domain: 10.0.2.3
Target entered: 10.0.2.3

Select a reconnaissance tool:
1) Whatweb
2) Nmap
3) DIRB
4) Exit
Enter your choice [1-4]: 1
Option 1
[+] Running WhatWeb against 10.0.2.3
^[[A^[[AERROR Opening: http://10.0.2.3 - execution expired
(icdfa@icdfa)-[~/lab3-recon]
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool

Use only against authorised lab targets.

Enter authorized target IP address or domain: 10.0.2.3/24
Target entered: 10.0.2.3/24

Select a reconnaissance tool:
1) Whatweb
2) Nmap
3) DIRB
4) Exit
Enter your choice [1-4]: 1
Option 1
[+] Running WhatWeb against 10.0.2.3/24
http://10.0.2.3/24 [404 Not Found] Apache[2.2.8], Country[RESERVED][ZZ], HTTPServer[Ubuntu Linux][Apache/2.2.8 (Ubuntu) DAV/2], IP[10.0.2.3], Title[404 Not Found], WebDAV[2]
(icdfa@icdfa)-[~/lab3-recon]
$
```

```
One-Machine2RuleAll (Snapshot 10) [Running] - Oracle VirtualBox
File Machine View Input Devices Help

icdfa@icdfa: ~/lab3-recon
File Actions Edit View Help

.2.8 (Ubuntu) DAV[2], IP[10.0.2.3], Title[404 Not Found], WebDAV[2]
(icdfa@icdfa)-[~/lab3-recon]
$ ./recon_tool.sh
ICDFA Beginner Reconnaissance Tool

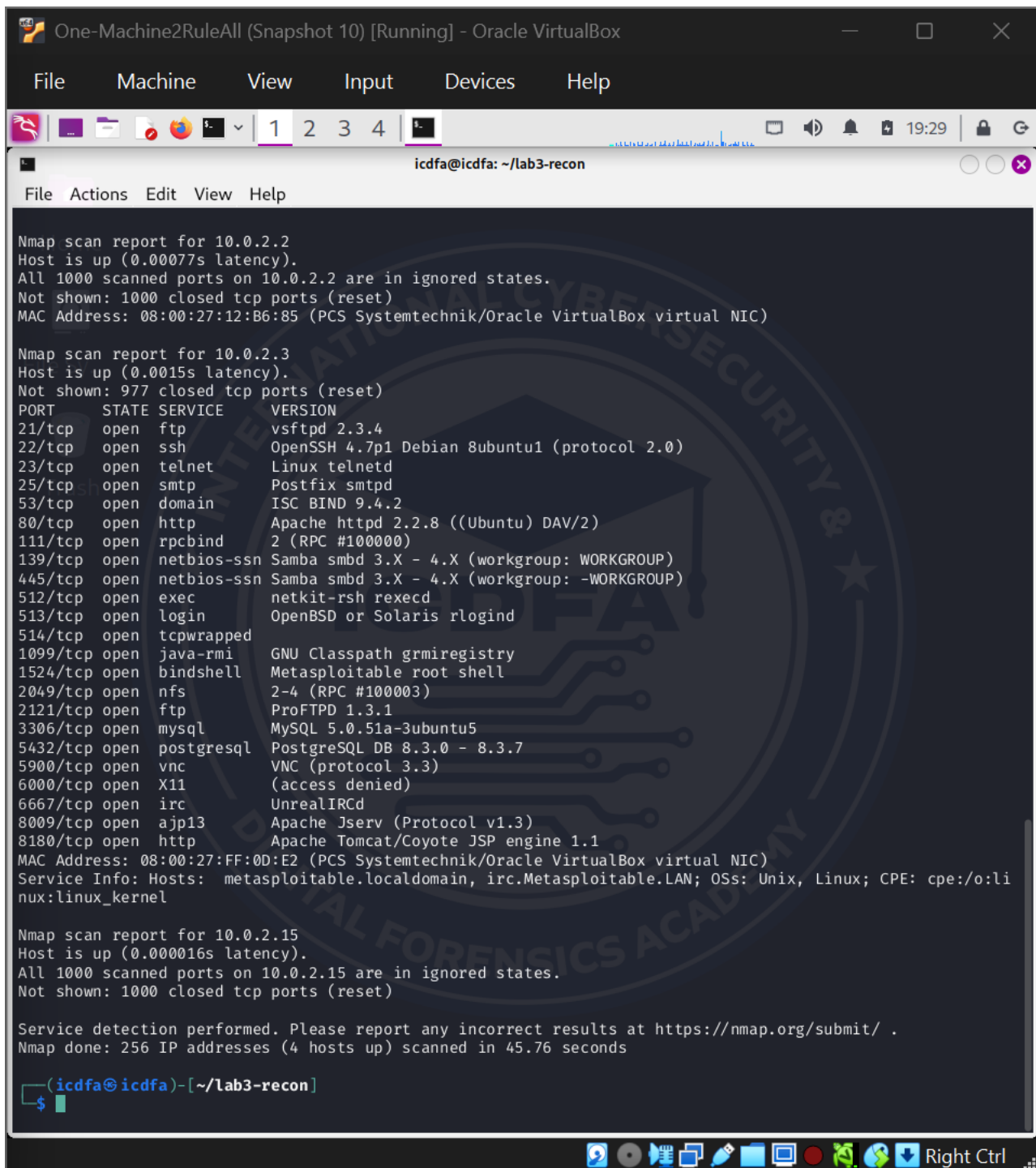
Use only against authorised lab targets.

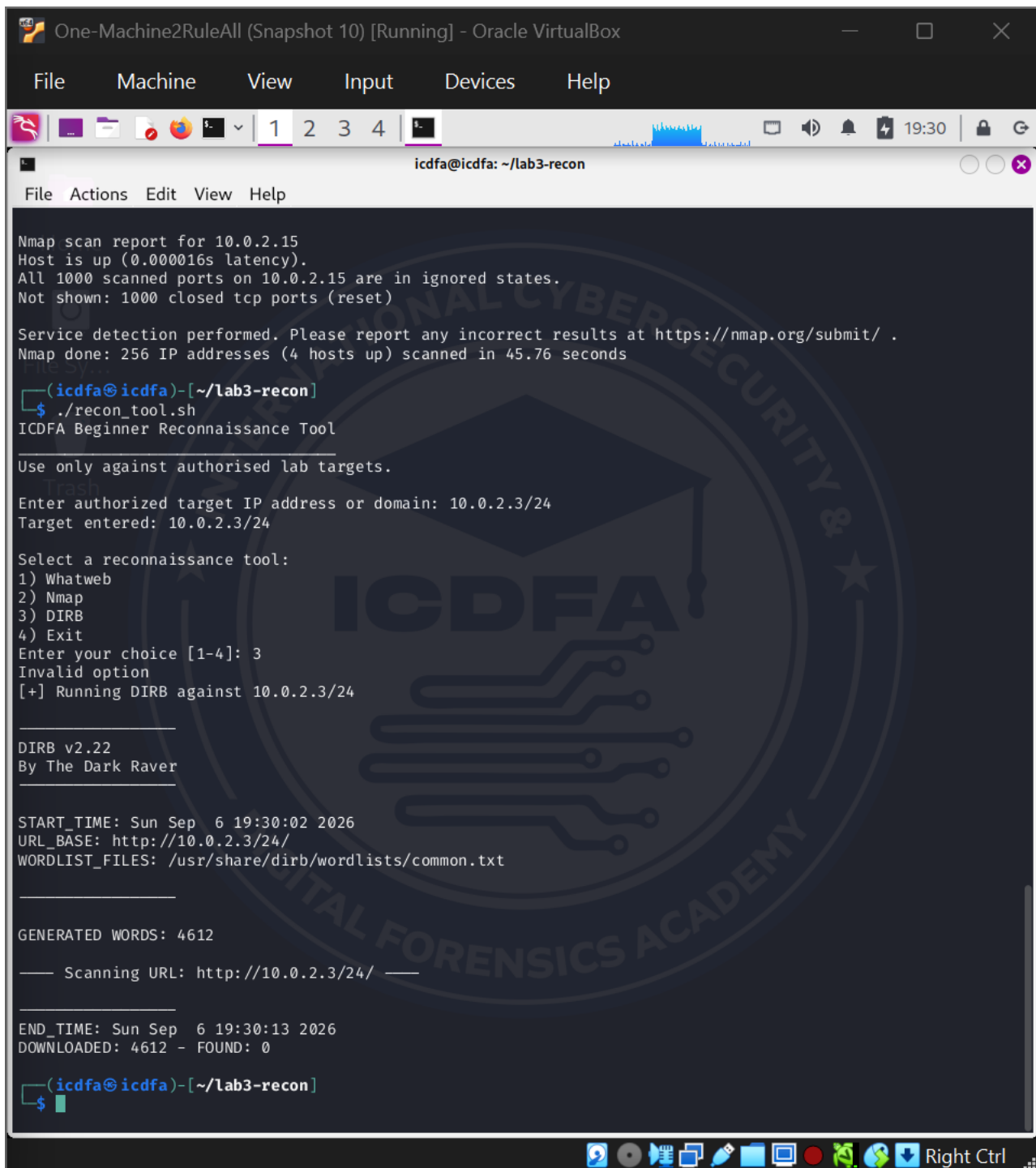
Enter authorized target IP address or domain: 10.0.2.3/24
Target entered: 10.0.2.3/24

Select a reconnaissance tool:
1) Whatweb
2) Nmap
3) DIRB
4) Exit
Enter your choice [1-4]: 2
Option 2
[+] Running Nmap service detection against 10.0.2.3/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-09-06 19:26 WAT
Stats: 0:00:01 elapsed; 0 hosts completed (0 up), 255 undergoing ARP Ping Scan
ARP Ping Scan Timing: About 54.51% done; ETC: 19:26 (0:00:01 remaining)
Stats: 0:00:28 elapsed; 252 hosts completed (3 up), 255 undergoing Host Discovery
Parallel DNS resolution of 1 host. Timing: About 0.00% done
Stats: 0:00:39 elapsed; 252 hosts completed (3 up), 3 undergoing Service Scan
Service scan Timing: About 40.74% done; ETC: 19:27 (0:00:09 remaining)
Nmap scan report for 10.0.2.1
Host is up (0.0018s latency).
Not shown: 996 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
135/tcp   open  msrpc        Microsoft Windows RPC
445/tcp   open  microsoft-ds?
2869/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
5985/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 52:55:0A:00:02:01 (Unknown)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Nmap scan report for 10.0.2.2
Host is up (0.00077s latency).
All 1000 scanned ports on 10.0.2.2 are in ignored states.
Not shown: 1000 closed tcp ports (reset)
MAC Address: 08:00:27:12:B6:85 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap scan report for 10.0.2.3
Host is up (0.0015s latency).
Not shown: 977 closed tcp ports (reset)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp          vsftpd 2.3.4
```





One-Machine2RuleAll (Snapshot 10) [Running] - Oracle VirtualBox

File Machine View Input Devices Help

1 2 3 4

icdfa@icdfa: ~/lab3-recon

```
File Actions Edit View Help
$ nano recon_tool.sh
(icdfa@icdfa)-[~/lab3-recon]
$ cat recon_tool.sh
#!/usr/bin/env bash

echo "ICDFA Beginner Reconnaissance Tool"
echo "_____ "
echo "Use only against authorised lab targets."

echo
read -rp "Enter authorized target IP address or domain: " target
echo "Target entered: $target"

if [[ -z "$target" ]]; then
    echo "Error: no target was entered."
    exit 1
fi

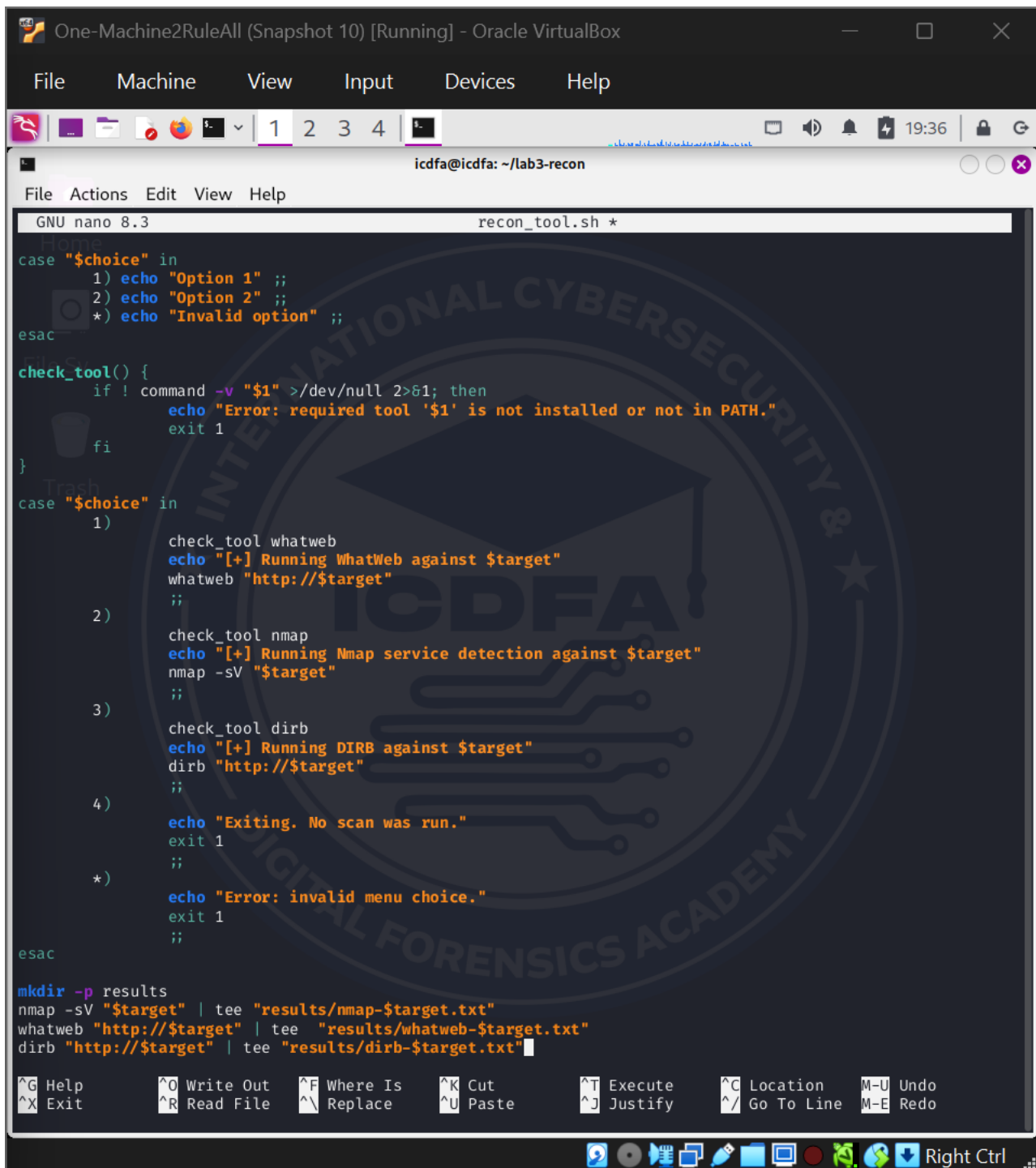
echo
echo "Select a reconnaissance tool:"
echo "1) Whatweb"
echo "2) Nmap"
echo "3) DIRB"
echo "4) Exit"
read -rp "Enter your choice [1-4]: " choice

case "$choice" in
    1) echo "Option 1" ;;
    2) echo "Option 2" ;;
    *) echo "Invalid option" ;;
esac

check_tool() {
    if ! command -v "$1" >/dev/null 2>&1; then
        echo "Error: required tool '$1' is not installed or not in PATH."
        exit 1
    fi
}

case "$choice" in
    1)
        check_tool whatweb
        echo "[+] Running WhatWeb against $target"
        whatweb "http://$target"
        ;;
    2)
        ;;
esac
```

Right Ctrl





Part 10

1. Shebang (`#!/usr/bin/env bash`): Specifies the absolute interpreter path used to execute the script instructions.
2. `read -rp`: `-r` prevents backslash escapes from being interpreted; `-p` prints a prompt string before waiting for input.
3. Quoting "\$target": Prevents word-splitting and glob expansion if the variable contains special characters or spaces.
4. `-z` test: Evaluates whether a string has zero length (checks if the input is empty).
5. `exit 1`: Terminates the script immediately with a standard general error status code.
6. `case` for menus: Provides clean pattern matching against multiple discrete options without nesting multiple `if/elif` blocks.
7. `;;` operator: Acts as a break terminator for a specific pattern block in a `case` statement.
8. `*` pattern: A default wildcard match that catches any input not matched by previous patterns.
9. `http://$target` in web tools: Explicitly defines the HTTP protocol and URI scheme required by URL-based scanners.
10. `nmap -sV`: Runs service and version detection on target open ports.
11. `command -v`: Verifies if a command binary is installed and executable within the current `$PATH`.
12. `chmod +x`: Adds execution permissions (`+x`) to the file's mode bits.
13. `./` syntax: Specifies that the target executable is located in the current working directory rather than system `$PATH`.
14. `tee` command: Duplicates standard input to both standard output (screen) and one or more files.

15. Hardcoding prevention: The read -rp prompt dynamically accepts user input into the target variable on each run.